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Position: On the Risks of Generative Engine Optimization in the Era of LLMs

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Abstract

Large language models (LLMs) are increasingly used as answer engines instead of ranked link lists in traditional web search. This shift enables generative engine optimization (GEO), in which advertisers, merchants, and GEO service providers tune content to influence what LLMs retrieve, cite, and recommend. This position paper argues that GEO turns LLM-based search into a new advertising and security surface. We systematize academic GEO work on recommendation manipulation and show that most studies share a narrow assumption: the optimized adversarial content is already present in the topk retrieval context, while differing in optimization methods, stealth constraints, and transferability. We then survey emerging commercial GEO providers and technical blogs to document how GEO is offered as a service that combines visibility tracking with LLM-guided content generation and cross-platform distribution on high authority websites. To unify these perspectives, we formalize GEO in RAG pipelines using a retrieval booster and ranking shifter message framework. We illustrate, with case studies, how adversarial edits to target content can shift LLM recommendations. Finally, we discuss implications for advertising practice, transparency, fairness, and defenses, and outline directions for measurement, regulation, and robust LLM design.

Keywords

computing and processing, generative engine optimization, knowledge corruption attacks, large language models, recommender systems, retrieval-augmented generation

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Index Terms—Generative Engine Optimization, Large Language Models, Retrieval-Augmented Generation, Knowledge Corruption Attacks, Document Augmentation Defense

I. INTRODUCTION

Search engine optimization (SEO) is the practice of improving the visibility of web pages in organic search through on-page and off-page techniques [1]. As search interfaces increasingly adopt large language models (LLMs), a related practice called generative engine optimization (GEO) has emerged. GEO gives content creators a way to shape how their material is retrieved, cited, and surfaced in generative responses [2]. Unlike SEO, which competes for ranked link positions, GEO focuses on inclusion and prominence within LLM-generated answers. It shifts competition from search result pages to answer space, meaning the limited area inside model outputs where citations and recommendations appear [2]. While

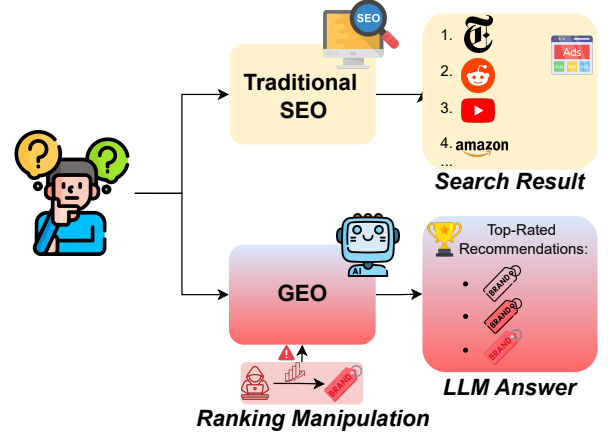


Fig. 1: Traditional SEO targets search result rankings, while GEO targets visibility inside LLM answers, enabling new forms of ranking manipulation.

several academic studies explore the GEO recommendation manipulation attack [3]–[6], a commercial GEO ecosystem is already forming. Companies such as ProFound [7], Goodie [8], and AirOps [9] offer GEO services that help clients monitor how AI-based answer engines respond to their brand’s keywords. These services extend familiar SEO practices to generative systems by tracking citations and advising creators on how to structure content so that it is more likely to appear in LLM-based search engine responses. This trend signals a broader shift toward practices that aim to influence LLM outputs, often without clear disclosure to end users.

Therefore, this shift raises several issues. First, the way LLMs prioritize content is not transparent. Since LLMs produce ranked or recommended outputs with limited attribu-

tion and unclear ranking signals, they are susceptible to manipulation. A prior work shows that strategically crafted text on product pages can influence model recommendations, raising concerns about fairness and robustness [3]. As shown in Fig.1, this non-transparency also creates opportunities for **paid visibility and citations** that lack disclosure and related patterns already exist today. Content farm networks pretend to be major news outlets and use lookalike domains to sell advertorial space at scale [10]. Some media companies, such as Red Ventures, have also been reported to use AI-generated and SEO-focused content to drive traffic and revenue [11]. These examples reveal a new attack interface for GEO, in which optimized content is created to dominate the evidence LLMs rely on for their recommendations.

Second, GEO poses risks to advertising and information integrity. Traditional advertising is governed by norms such as the FTC requirement that native ads be disclosed as promotional content [12]. GEO bypasses these norms by embedding engineered visibility directly into LLM outputs without transparent labeling. This blurs the boundary between editorial content and advertising and creates a regulatory grey zone, where influence can be purchased through citations and prompt steering.

Third, as major LLM-based search engines adopt retrieval-augmented generation (RAG) pipelines for online search [13]–[16], they inherit vulnerabilities from retrieval systems. Because the RAG pipeline conditions on external documents, the whole LLM-based search engine system is exposed to biased or misleading inputs and can produce hallucinated responses. Additionally, optimized GEO content may itself become a source of such biased material.

If GEO practices continue to develop without oversight, actors with greater technical and financial resources may dominate LLM-mediated information spaces, sidelining less resourced yet more credible sources [2], [3]. This dynamic threatens the promise that LLMs will provide accurate, relatively unbiased, high-integrity information.

In this position paper, we argue that LLM-based answer engines transform GEO from a niche marketing tactic into a systemic risk for advertising transparency, content fairness, and RAG-based answer systems. Our contributions are as follows.

- We provide evidence that GEO and its recommendation manipulation attack are already practiced in both academia and industry, rather than remaining a speculative concern.
- We formalize the GEO recommendation manipulation attack in RAG-based LLM search engines as a bi-level optimization problem with retrieval booster and ranking shifter messages, capturing how adversarial content can bias both retrieval and ranking.
- We characterize how GEO recommendation manipulation attacks are implemented in current academic work, including the typical optimization targets, experimental designs, and evaluation setups.

- We survey real-world companies to document how GEO and the recommendation manipulation attack are implemented as a commercial service for clients.
- We present a comparative analysis of academic and industry GEO-related practices, summarized in a table that highlights their objectives, methods, and evaluation strategies.

Taken together, our analysis links academic prototypes of GEO with emerging commercial services and highlights where their practices align and where they diverge. We argue that these real-world developments show that GEO already poses practical, not just hypothetical, risks. This underscores the need for safeguards that protect transparency, fairness, and information integrity as LLM-based answer engines expand.

II. BACKGROUND

Retrieval-Augmented Generation (RAG) with LLMs: RAG-based LLM systems combine a retriever and a generator to reformulate queries, select relevant sources, and produce attributed answers [17]. Recent work uses RAG to strengthen LLMs on tasks that require current and reliable knowledge, including question answering, scientific applications, and recommendation [18], [19]. Because RAG relies on an external and editable document collection, the evidence it selects can be influenced [20]. Small changes to product pages or related articles can shift what the system retrieves and which brands it recommends. This flexibility is useful for personalization, but it also creates opportunities for manipulation through crafted prompts or content edits.

Adversarial Attacks with Prompt Injection: Prompt injection attacks involve planting adversarial instructions within the input to an LLM, to steer the model toward generating responses that align with the attacker’s objectives [21], [22]. Recent research shows that RAG-based LLM systems are vulnerable to strategic manipulation [23], especially in e-commerce settings. Kumar and Lakkaraju [3] find that inserting a carefully optimized text sequence into product descriptions can significantly raise a product’s rank in generative recommendations, illustrating how adversarial changes can distort fair competition. Pfrommer *et al.* [5] formalize ranking manipulation attacks on conversational search engines and show that prompt injection and adversarial rephrasing can consistently push low-ranked items higher in LLM-generated results. Extending this line of work, Nestaas *et al.* [6] introduce preference manipulation attacks that exploit model conditioning to bias answer selection across search and plugin ecosystems. Together, these studies show that adversarial prompts and retrieval signals can effectively steer model behavior, revealing both the fragility of current alignment mechanisms and the potential for hidden commercial influence.

SEO and GEO: Traditional SEO seeks to improve visibility in ranked search results through on-page optimization and authority building strategies. With the rise of generative search engines, Aggarwal *et al.* [2] introduce GEO, a framework that helps content creators improve their visibility in generative outputs. GEO extends traditional SEO by optimizing content

for inclusion in LLM-generated answers rather than for hyperlink rankings. Specifically, GEO works by influencing the data that LLMs retrieve and cite. When an assistant provides a list of “recommended moisturizers,” it often attaches resource URLs linking to product descriptions, ingredient reviews, or blog posts. These URLs become strategic intervention points. By optimizing content for perceived authority through search optimization, domain reputation, and keyword tuning, brands can increase the likelihood that their pages are retrieved and cited. The LLM’s answer then indirectly functions as a persuasive advertisement framed as expert advice. Chen *et al.* [24] show that AI search engines tend to favor authoritative, citation rich, and well structured sources. This tendency can be strategically exploited to gain visibility.

III. MOTIVATION

Figure 2 shows how the product recommendation landscape differs between traditional SEO and GEO. When a user searches for “best skincare products for moisture” on Google, the top results are dominated by paid placements from premium brands such as Lancôme, La Mer, and RéVive. These are labeled as “Sponsored products” and are sorted through advertising bids, where visibility is determined by marketing budgets.

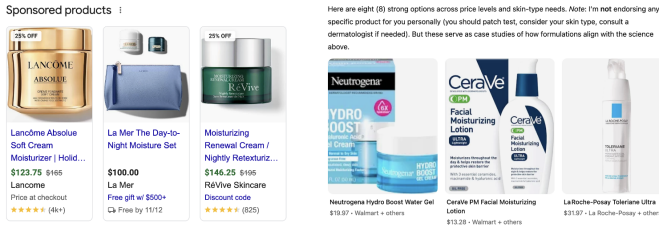


Fig. 2: Comparison between SEO-based product results (left) and GEO-influenced LLM recommendations (right) for the skincare query.

In contrast, an LLM-based search engine, such as ChatGPT, produces recommendations that emphasize ingredient quality and dermatological science. Instead of promoting luxury brands, it suggests products like CeraVe or Neutrogena, highlighting key hydration mechanisms such as hyaluronic acid or ceramide content. This difference shows that traditional search rankings reflect advertising and SEO, while LLM-based recommendations reflect what appears valuable in the model’s training or retrieval data.

This shift creates both opportunities and risks. LLM-based search engines can broaden visibility by highlighting affordable products that have stronger scientific support, rather than only those backed by large advertising budgets. At the same time, it creates a new channel for commercial optimization, where brands no longer compete for advertisement slots but instead compete for inclusion in the LLM’s retrieved evidence, reasoning, and citations [2], [3], [6].

Motivation. Despite these developments, GEO on LLM-based search engines is not yet well surveyed in either academia

or industry. Existing research examines specific mechanisms and settings in isolation [2]–[6], and there is no unified view of threat surface, attacker’s goals, capabilities, optimization methods, evaluation methodology, and practical impact. On the industry side, commercial providers describe GEO at a high level in marketing and technical blogs, but reveal little about implementation details or defenses.

In this paper, we address this gap by systematizing known academic GEO and its recommendation manipulation attacks, comparing them with emerging commercial practices. We also empirically study how the GEO adversarial edits to the target content can shift LLM recommendations.

IV. THREAT MODEL

Prior GEO recommendation manipulation attacks model attackers as entities that seek to influence recommendations in LLM-based answer engines. The attacker can be either (i) **merchants** acting on their own behalf, or (ii) **third-party GEO service providers** operating on behalf of one or more merchants. In both cases, the attacker aims to influence which target products and brands appear, how they are ranked, and how they are described in LLM-based answer engine responses.

A. Attacker capabilities.

Existing work considers both *white-box* and *black-box* settings. In the *white-box* case, the attacker has access to model parameters and can optimize over input tokens directly. In the *black-box* case, the attacker can query one or more LLM-based search engines and observe their responses, but has no access to parameters, gradients, or internal retrieval indices.

Across these settings, the attacker can modify or create adversarial content in the following ways:

- The attacker can edit product pages and other content on the merchant’s own web domains (e.g., catalog pages, blogs, and help centers).
- The attacker can generate and post crafted adversarial content on third-party platforms that are likely to be crawled, retrieved, or trusted by LLM systems, such as forums, Q&A sites, review sites, and knowledge bases.
- The attacker can adjust content over time based on observed LLM-based search engine responses, using them as feedback on which edits increase or decrease visibility.

We refer to any text created or modified by the attacker to influence the LLM’s responses as *adversarial content*.

B. Attacker’s goals.

The attacker’s primary goal is to bias the recommendations produced from LLM-based search engines. At the item level, the attacker may:

- **Promote** specific products or brands, increasing their rank, exposure, or likelihood of being named as a “top” or “recommended” choice.
- **Demote** competing products, reducing their rank, exposure, or positive mentions in the same responses.

V. METHODOLOGY

We constructed 2 complementary corpora: an *academic corpus* and an *industry corpus*. Both focused on GEO and adjacent practices such as optimizing content for AI driven answers, and both were restricted to sources that (i) provided specific and verifiable claims, (ii) had identifiable and reputable provenance (peer reviewed venues, recognized experts, established agencies, or official company pages), and (iii) demonstrated recency and visibility (*e.g.*, search ranking, shares, or cross references).

A. Academic corpus.

For academic work, we searched scholarly databases and search engines (Google Scholar, Semantic Scholar, and publisher portals) using time filters and varied queries related to GEO, LLM answer optimization, and retrieval augmented generation. We supplemented these searches with forward and backward citation chaining from key papers, as well as curated research newsletters. We included peer-reviewed papers, preprints from recognized research groups, and institutional reports that met our inclusion criteria.

B. Industry corpus.

For industry and commercial materials, we combined Google and Bing searches with time filters and diverse queries, monitored curated newsletters and major industry blogs, and reviewed practitioner forums and communities. We conducted a targeted review of company websites offering GEO-related services or guidance and, for one provider, interviewed with the sales team to obtain a more detailed view of their optimization process. The corpus includes blogs, product pages, and agency guidance that met our inclusion criteria.

VI. ANALYSIS OF GEO OPTIMIZATION PRACTICES

Drawing on the literature and observations of industry practice, we formalize a common system architecture and describe how GEO can intervene in LLM-based search engine pipelines to bias the final recommendation response.

A. System architecture

Figure 3 illustrates a three-block framework: from left to right (i) *LLMs Block*: the user makes a query to LLM and gets the recommendation; and (ii) *Search Flow Block*: The LLM gathers information by crawling authoritative platforms such as Wikipedia, Yelp, Reddit, and Amazon. The LLM then builds indexes of the collected content. After that, the LLM ranks the indexed information to identify the most relevant pieces, which is also called top- k RAG contexts for LLM. (iii) *Generative Engine Optimization Block*: an attacker manipulates the LLM’s output by injecting optimized adversarial content into the LLM’s search ecosystem. The attacker can either **(a) optimizes a merchant website so it aligns with features that LLMs prefer**, such as structured titles, SEO-style keywords, and authoritative formatting. This increases the chance that the LLM will index and rank the site as a high-quality source. The

attacker can also **(b) amplify the target topic by crafting optimized review blogs on popular, high-authority platforms that LLMs frequently retrieve from and cite**, for example, posting positive review blogs, high-engagement comments, and promotional content across forums, social media, and review sites. These coordinated actions raise the visibility and ranking of attacker-crafted adversarial information, making it more likely to appear in the top- k retrieval context. As a result, the LLM-based search engines are more likely to treat these sources as credible evidence and incorporate them into their summaries and recommendations.

Together, in this example, a user asks, “*Heading to LA soon! What are the best things to do?*” The LLM processes the adversarial context searched from different online platforms and elevates the attacker’s target item, *French Dip Sandwich*, as the top recommendation. In the following section, we will focus on the technical details of the *Generative Engine Optimization Block*.

B. Technical implementation.

Given a target topic t , users issue paraphrased queries $q \sim \Pi(t)$ to an LLM-based search engine. Each query triggers a RAG-style retrieval process that returns the top- k relevant webpages as context. The LLM component then generates a response grounded in this context, which can be biased if adversarial content is present.

To maximize visibility and ranking impact on the LLM’s outputs, a GEO recommendation manipulation attack can be modeled as a bi-level optimization problem. Inspired by PoisonedRAG [23], an attacker constructs a *retrieval booster message* b , optimized for retrievability, and a *ranking shifter message* c , optimized for rank-shift impact. The adversarial content b and c can be jointly synthesized and deployed across both merchant websites and topic-related external websites.

a) Retrieval booster message (b): Following prior work [23], a retrieval booster message b is a passage edited to improve its retrievability across paraphrased queries. This involves adjusting titles, headings, anchors, metadata, keyword coverage, and backlinks, as well as generating diverse paraphrased variants, so that b consistently appears in the top- k retrieved results. A length constraint keeps b realistic. We denote the retrieval-boost objective by $J_{\text{boost}}(b)$ and choose b to solve

$$\max_b J_{\text{boost}}(b) = \mathbb{E}_{q \sim \Pi(t)} [\text{Sim}(q, b)] \quad \text{s.t.} \quad \ell(b) \leq L,$$

where $\text{Sim}(q, b)$ is the fused similarity score between query q and retrieval booster message b , and $\ell(b) \leq L$ constrains the length of the retrieval booster message.

b) Ranking shifter message (c): Following prior work on GEO [2], a ranking shifter message c changes how the LLM ranks and describes the target topic t once it is included in the retrieved context. Ranking shifter message c is a constrained LLM-crafted passage that is fluent, on topic, and verifiable. Incorporating statistics, citations, and domain-specific terminology in c has been reported to improve LLM visibility,

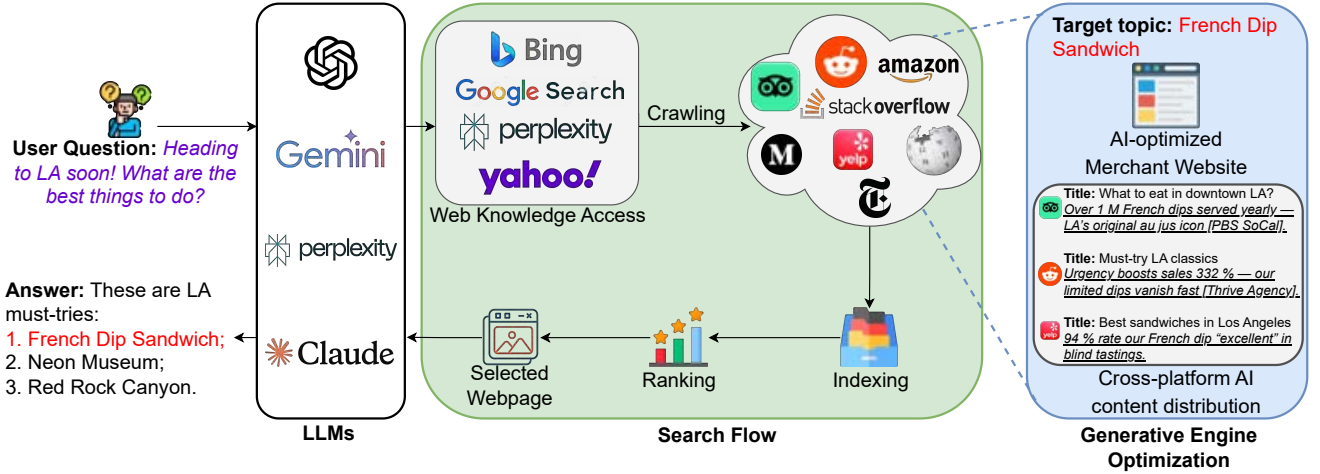


Fig. 3: Overview of the GEO recommendation manipulation attack. Given a target topic and a promote or demote objective, attackers can manipulate target websites and inject crafted adversarial content into high-authority platforms to increase or decrease the topic’s visibility to LLM-based search engines.

Method	Assumption	Optimization Method	Injection Position	Attacker’s Goal	Evaluation Setup	Stealth Constraints	Attack Transferability
GEO [2]	Adversarial content in the top- k retrieval context	LLM-based rewriting	Rewriting	Promotion	Offline	None	Single Model
LLM Visibility [3]		GCG	Appending	Promotion	Offline	None	Single Model
Stealthy LLM [4]		LLM-based rewriting	Insertion	Promotion & demotion	Offline	Token budget + semantic similarity	2 models
Ranking Manipulation [5]		TAP	Prepending	Promotion	Offline	Manual hide	4 models
Adversarial Search [6]		Manually crafted text	Appending	Promotion	Online	Manual hide	3 models

TABLE I: Comparison of academic LLM-based recommendation manipulation attacks.

allowing such passages to subtly promote or demote the topic t in the final answer.

Let $C(q)$ denote the retrieved context that the LLM uses when answering query q :

$$C(q) \subseteq \text{Top-}k_R(q; \mathcal{D} \cup \{b, c\}),$$

where $\mathcal{D}$ is the clean corpus from the Internet. The operator $\text{Top-}k_R(q; \cdot)$ returns the k highest ranked documents for query q according to the retrieval model R .

Downstream manipulation is quantified by a utility function U that measures promotion or demotion of the target topic t given a user query q , typically via the change in its rank. We denote the ranking shifter objective by $J_{\text{shift}}(c)$ and choose c to solve

$$J_{\text{shift}}(c) = \mathbb{E}_{q \sim \Pi(t)} [U(q, t; C(q))].$$

For *promotion* attacks, c is chosen to maximize this objective,

$$c_{\text{promote}}^* = \arg \max_c J_{\text{shift}}(c),$$

whereas for *demotion* attacks, c is chosen to minimize it,

$$c_{\text{demote}}^* = \arg \min_c J_{\text{shift}}(c).$$

GEO pipelines iteratively tune b for retrievability and c for downstream ranking impact. After each crawl and indexing cycle, b can be updated using retrieval logs, and c can be

updated using end-to-end exposure and ranking metrics. These updates continue until the desired performance thresholds are met.

VII. EMPIRICAL OBSERVATION

GEO, as a new marketing channel, remains underexplored, leaving marketing professionals without a shared understanding of its best practices. To capture the evolving interpretations and applications of GEO, we examined both academic and industry sources, including peer-reviewed papers, blogs, industry reports, and company publications. This section presents our observations of GEO-related materials and services across academic and industrial contexts.

A. Academic Observation

Since 2023, starting with GEO [2], researchers have begun to systematically study how adversaries can manipulate LLM’s recommendation responses under the e-commerce settings by altering retrieved content $C(q)$. Subsequent works [3]–[6] extend this line of research. Table I summarizes existing GEO recommendation manipulation attacks along the following dimensions: assumption, optimization method, injection position, attacker goal, evaluation setup, stealth constraints, and attack transferability.

For assumptions, we find that all surveyed academic GEO recommendation manipulation studies rely on the same idea:

they assume the adversarial content b and c is already included in the top- k retrieval context $C(q)$, in other words, they assume the adversarial content has already been retrieved. Under this assumption, the attacker only needs to optimize the ranking shift objective $J_{\text{shift}}(c)$ and can leave the retrieval booster message b untouched. As a result, these studies focus on changing rankings based on the retrieved data, rather than on influencing the retrieval pipeline itself [2]–[6].

Optimization methods specify how adversarial content is generated. They range from LLM-based rewriting and Tree of Attacks with Pruning (TAP) [2], [4], [5] to a white-box gradient attack [3] and manually written prompts [6]. The placement of adversarial text also differs: GEO [2] rewrites the entire page, while other work appends [3], [6] or prepends [5] blocks of text, or inserts in-line edits [4]. Almost all attacks focus solely on promotion and maximizing $J_{\text{shift}}(c)$, with only one study treating both promotion and demotion of target items [4]. In other words, it can either choose to maximize or minimize $J_{\text{shift}}(c)$.

Evaluations are dominated by controlled offline setups [2]–[5], where models are queried on static datasets rather than through live search engines. Only one study probes production systems by issuing queries directly to deployed LLM-based search interfaces [6], but in the limited search domain *e.g.* *spylab.ai*. Attack stealthiness is addressed in a limited way: two related work [5], [6] hide adversarial text manually, for example, by instructing the LLM not to mention the injected source or by embedding it in HTML elements. Nazary *et al.* [4] further quantify stealth by limiting the adversarial token budget and enforcing high semantic similarity to the original text.

Attack transferability measures how well a manipulation generalizes beyond the specific model or configuration it was optimized for, but it is only tested on small model sets [2]–[6]. 2 methods are evaluated on a single model [2], [3], 1 on 2 models [4], 1 on 4 models [5], and 1 on 3 production systems [6]. Taken together, these patterns suggest that current research covers several attack dimensions but relies on strong assumptions and offers only a shallow treatment of retrieval, stealth, and cross-model robustness.

B. Industry Observation

Beyond the academic literature, GEO is already an active commercial practice. Startups such as Goodie [8], AthenaHQ [25], AirOps [9], and ProFound [7] explicitly market services to improve visibility in LLM-based search engines. Several reports of multi-million dollar seed and Series A or B rounds [26]–[30], suggesting that investors view GEO as a distinct and commercially valuable capability.

Unlike academic work, which describes models and evaluation protocols in detail, these firms mostly communicate through product pages and high-level technical blogs that reveal little about implementation. They also target different surfaces: instead of only rewriting client websites, they report crafting optimized adversarial posts and distributing them on different platforms such as Reddit and Wikipedia. To understand these practices, we examine practitioner sources,

including company sites and technical blogs, to characterize how GEO is currently implemented in industry.

1) *Companies*: Fig. 4 illustrates how Goodie [8], AirOps [9], and ProFound [7] track visibility, citations, and rankings in LLM-based search engines, and generate reports that reveal which queries, pages, snippets, and phrasings are most frequently cited and highly ranked.

Based on these reports and client keywords, some providers first optimize the retrieval booster message b by prompting an LLM to generate different paraphrased candidate user queries $\hat{q}$ to increase $J_{\text{boost}}(b)$. For each $\hat{q}$, they then prompt the LLM again to generate adversarial content that acts as a ranking shifter c , tuned to increase or decrease the visibility of the target in terms of $J_{\text{shift}}(c)$.

The resulting optimized adversarial content is then distributed on external websites such as Reddit, Wikipedia, forums, and Q&A sites, as well as on clients’ own blogs and help centers. From our observations, providers do sometimes adjust a client’s official site, but these edits resemble traditional SEO, and most effort focuses on adversarial content generation and cross-platform distribution.

2) *Technical Blogs*: CapGo AI [31] presents GEO as an extension of SEO and programmatic SEO, emphasizing content tailored for AI assistants so that brands can be recommended directly in responses rather than only linking to web pages. SaleSmartly [32] argues that the October 2024 launch of ChatGPT Search shifts optimization from traditional rankings toward intent-aligned, well-structured content. Therefore, the content can be easily retrieved, summarized, and quoted inside ChatGPT. OpenAI’s Merchants page [33] outlines product discovery flows inside ChatGPT using rich item cards and merchant listings, highlighting a new interaction surface where GEO-styled optimization of product metadata and descriptions may directly affect which items are shown to users.

VIII. COMPARISON BETWEEN ACADEMIC GEO FRAMEWORKS AND COMMERCIAL IMPLEMENTATIONS

This section compares academic GEO frameworks and commercial implementations along both technical and service dimensions. Table II contrasts research prototypes and deployed systems in terms of target LLM knowledge, threat surface, search domain, optimization metrics, how these metrics are obtained, optimization methods, and datasets. Table III then summarizes service-level characteristics that matter to adopters, including which LLMs are targeted, expected delivery time, the form of delivered recommendations, and the main external platforms where content is placed.

a) *Commonalities*.: Across both academic and commercial GEO, the main thing that changes is the content. They change the text that the LLM-based search engines crawl and retrieve, rather than model architectures or retriever parameters. Academic work rewrites product descriptions or synthetic webpages and then measures how rankings and visibility scores change. Commercial tools generate or rewrite blogs, reviews, Q&A entries, and place them on client sites and high authority external websites. In addition, most academic

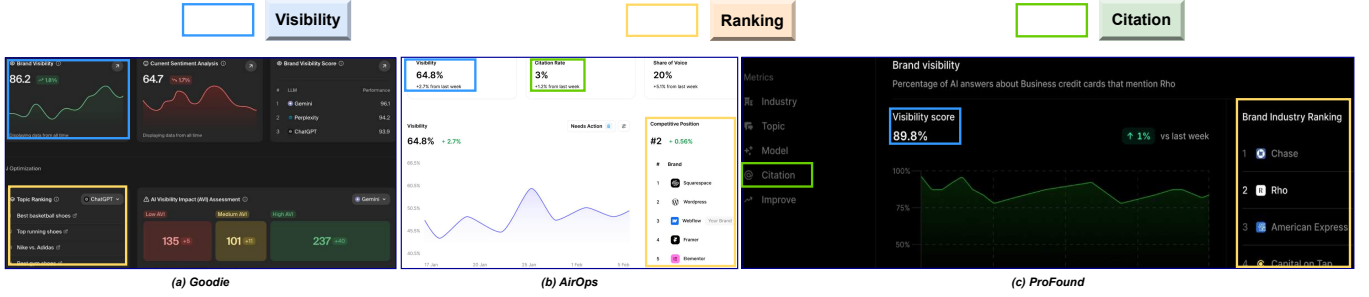


Fig. 4: Comparison of Goodie, AirOps, and ProFound

Domain	Method	Target LLM Knowledge	Threat Surface	Search Domain	Metric to Optimize	How to Get Metric	Service Type	Optimization Method	Dataset
Academia	LLM Visibility [3]	White-box	Merchant's Website	Offline	Product Rank	Offline web query	N/A	GCG Attack	Fictitious Catalog
	GEO [2]	Position-adjusted word count, G-EVAL Metrics			LLM-based rewriting			G-Bench	
	Stealthy LLM [4]	Recall@k, nDCG@k			LLM-based rewriting			MovieLens	
	Ranking Manipulation [5]	Ranking Score			TAP			RAGDOLL	
	Adversarial Search [6]	Recommendation Rate, Citation Rate	Manual Crafted	Dummy Websites					
Industry	ProFound [7]	Black-box	Merchant's Website & External Websites	Online	Visibility Score, Ranking, Citation Score	Online web query	API Provider & Service Provider	Visibility Score Guided LLM Generation: – Query Expansion – Query-Driven Content Generation – Citation-Oriented Content Distribution	N/A
	Goodie [8]					Sourced via API provider	Service Provider		
	AirOps [9]		External Websites						
	AthenaHQ [25]								
	WOYAOBIAO Software				External Websites			Ranking	

TABLE II: Academic and commercial approaches to GEO-style optimization for LLM-based search engines. The table compares methods by domain, target LLM knowledge, threat surface, search domain, optimization metric, how it is obtained, service type, optimization method, and dataset settings.

and commercial systems treat the target LLM as a black box, with no access to model parameters and only query–response behavior available for optimization.

b) Threat surface and search domain.: Academic attacks mostly change content on merchant websites and assume that the optimized adversarial content is already included in the top- k retrieved context. Industry faces a broader and less predictable surface. Providers must work with real-world search pipelines and increase content retrievability by generating multiple paraphrases of user queries $\hat{q}$. Then, they modify client sites and also distribute optimized adversarial content to high-authority external websites (*e.g.*, Wikipedia, Reddit, Yelp, Medium) that LLMs are likely to crawl, retrieve from, and cite.

c) Environment and data.: Academic works optimize and evaluate attacks on fixed offline datasets, such as fictitious catalogs, MovieLens [34], RAGDOLL, or dummy websites. Commercial systems instead optimize and validate directly on the open web and live LLM products, using whatever content current search pipelines crawl and retrieve. This difference in data sources makes academic results easy to reproduce, but commercial procedures harder to analyze from the outside.

d) Optimization and metrics.: Research prototypes focus on clear algorithms and metrics. They use explicit optimization methods (such as GCG, LLM-based rewriting, or TAP), well-defined evaluation metrics (rank, Recall@k, nDCG@k, visibility score). Commercials typically prompt LLM to generate optimized adversarial content and focus on outcome metrics

such as visibility score, being mentioned in answers, or achieving a top-5 ranking, which are directly tied to downstream revenue. They obtain these metrics through repeated online queries or API signals, and often combine heuristic rules with LLM-guided content generation and distribution, tracking changes over time. For example, a GEO provider working with a skincare brand may track how often the brand is named or cited when users ask for “best moisturizers” across ChatGPT, Perplexity, and Google AI Overview, then iteratively adjust blog posts, reviews, and Wikipedia style entries until the brand appears in the top answer panel and reported click-through and sales metrics increase.

e) Service model and constraints.: As shown in Table III, academic work does not sell services. It releases code, datasets, and protocols. Commercial GEO turns optimization into a product. Providers specify which LLM families they target, promise delivery time windows (from days to months), and sell concrete outcomes, such as being cited in answer panels or reaching a top- k position on specific platforms.

Academic GEO frameworks and commercial implementations are closely related but operate in different regimes. Academic work studies GEO under controlled offline assumptions, with explicit threat surfaces, optimization methods, and public datasets that make mechanisms and risks transparent and reproducible. Commercial GEO applies similar ideas but over a different threat surface and in black-box, outcome-driven pipelines on the open web, targeting specific LLMs, delivery windows, and business metrics such as mentions and top- k

Company	Target LLM	Delivery Time	Delivered Service	External Websites
ProFound [7]	ChatGPT, Gemini, Google AI Overviews, Perplexity, Claude, Grok, Deepseek	1-3 months	Mentioned in the LLM response	Wikipedia, Reddit, X, Medium, Yelp, Google Reviews, Reuters, Quora etc
Goodie [8]				
AirOps [9]				
AthenaHQ [25]				
WOYAOBIAO Software	Doubao, Qwen Deepseek, Ernie, Yuanbao	1-10 days	Top-5 ranking	RedNote, Douyin, Toutiao, NetEase, Sohu, Zhihu, Jianshu, etc

TABLE III: Summary of companies offering GEO service, showing the target LLMs, estimated delivery time to achieve results, delivered service type, and main content platforms targeted.

rankings in LLM-based responses. Together, these perspectives show that GEO is both a concrete engineering practice and a growing security risk, with academic models providing conceptual tools and testbeds, and industry deployments revealing feasibility, cost, and real-world impact.

IX. DISCUSSION AND IMPLICATIONS

This section examines the broader implications of GEO-style recommendation manipulation attacks. It considers effects on advertising, visibility in LLM-based search engines, fairness, and user trust, and highlights conflicting incentives and open questions for measurement and defense.

A. Commercial and advertising implications

GEO shifts advertising from buying visible ad slots to shaping how LLMs formulate answers [2], [3]. Advertisers optimize upstream content so that brands occupy scarce “answer real estate” such as citations, recommendations, and narrative framing. This blurs the line between advice and promotion and puts pressure on existing native advertising disclosure norms [12]. Competition also moves from impressions and clicks to measures such as share of answer and share of citations. This change incentivizes companies with significant financial and technical resources to invest in continuous GEO tactics and pushes platforms to monetize answer panels in ways that users may struggle to distinguish from organic content.

A further concern is the competitive dynamic that GEO can create. Preference manipulation attacks can produce a prisoner’s dilemma, where each brand gains by adopting aggressive GEO tactics, but if many competitors do the same, all of them lose visibility overall and the quality of answers degrades for users [6]. In this setting, firms feel pressure to attack simply to avoid falling behind, even when the collective outcome is worse for everyone.

B. Transparency and accountability

Transparency is central to trust, yet current LLM systems provide limited insight into how answers are constructed. Users may see a few cited documents but rarely understand why those sources were selected or how they influenced the final response [17], [18]. Without clear records, it is difficult to determine whether an answer relied on reliable evidence or on GEO-optimized materials.

Future systems should include mechanisms for tracking and exposing provenance, including which documents were retrieved, how they were weighted, and whether any content was paid or promotional. Lightweight standards, such as metadata files indicating whether website content may be reused by AI and logs of retrieved and cited sources, could make transparency measurable and auditable and provide a basis for regulatory oversight.

C. Content fairness and user trust

GEO also raises fairness concerns. Companies and organizations with greater financial and technical resources can systematically tune their content for visibility, while smaller or independent voices may struggle to compete [5], [19]. This risks reproducing, and potentially amplifying, provider biases already observed in online search within generative systems [24], [35]. If the same few sources dominate answers, users receive a narrower range of perspectives. Mitigation requires mechanisms to monitor and balance exposure, for example by tracking source diversity, limiting repeated use of heavily optimized content, and explicitly incorporating diversity alongside relevance in generative ranking objectives.

User trust depends on fairness, but also on honesty and clarity. Many people implicitly assume that LLM answers are neutral and factual [20]. When commercial content is mixed into fluent explanations without disclosure, the persuasive force of natural language can covertly shape attitudes [36]. If users later discover that ostensibly neutral recommendations were optimized for sales or exposure, they may feel deceived and reduce their reliance on the system. Clear labeling of sponsored or optimized material is therefore crucial for preserving credibility and avoiding public backlash.

X. FUTURE WORK

Future work should develop defenses and auditing tools for GEO, and identify transparent practices that improve visibility without harming fairness, user trust, or content quality.

A. Implications for advertisers and merchants

Conducting GEO on the open web means accepting that retrieval and ranking are black boxes and that the environment is noisy. Instead of trying to control the whole search flow, future work can use experimental designs. One option is to run randomized interventions on matched products or topics, where only some items receive a GEO treatment (*e.g.*, posting on specific external websites or changing citation style) and others act as controls. Researchers can then repeatedly query target LLM engines over time and log which brands are mentioned, which URLs are cited, and at what rank they appear before, during, and after the intervention. This kind of time series logging (a longitudinal design) makes it possible to observe how visibility changes as GEO tactics are applied.

To estimate causal effects, visibility changes for treated items can be compared to changes for control items. One simple approach is to compute, for each group, the difference between visibility before and after the intervention, and then

take the difference between these two changes. This difference in differences isolates the extra gain that is likely due to GEO, rather than to unrelated changes in the underlying LLM or search pipeline.

Second, researchers and policymakers can develop concrete ethical and regulatory frameworks for LLM advertising. This can include drafting and testing codes of conduct for GEO agencies and proposing disclosure rules for optimized content inside answers. They can also run user studies to measure how different labels or explanation formats affect trust, perceived bias, and willingness to follow recommendations.

B. Defensive strategies for LLMs

GEO exposes open problems in robust retrieval and ranking for LLM-based systems. One line of work is to design detectors and filters that identify strategic text sequences and adversarial context patterns of the kind used to shift rankings or recommendations [3], [5], [6]. These mechanisms could operate at indexing time, for example by scoring documents for suspicious optimization signals, or at inference time by downweighting or masking context segments that resemble prompt injections.

A second line is architectural. RAG and conversational search pipelines can be modified to reduce the influence of any single document, enforce source diversity, and incorporate provenance and reputation signals, so that manipulative pages cannot easily dominate the top- k context [2], [5].

A third direction is to build benchmarks and metrics that explicitly capture robustness to visibility attacks, for example, by extending GEO style evaluations with adversarial product catalogs or dummy sites [2], [6]. Such benchmarks would let developers compare defenses, track progress over time, and reason about trade-offs between answer quality, fairness, and resistance to commercial manipulation.

XI. CONCLUSION

GEO shifts optimization from search result rankings to the answer space of LLM-based search engine systems. Instead of competing only for higher positions in a list of links, actors try to influence which sources LLMs retrieve, cite, and recommend inside ostensibly neutral responses, creating new risks for transparency, fairness, and RAG-based recommendation systems.

This position paper shows that GEO manipulation is already practiced in both academia and industry. It analyzes how academic attacks are implemented and evaluated, and documents how companies offer GEO as a commercial service. It also compares these practices across objectives, methods, and evaluation strategies.

These findings suggest that GEO should be treated as a central topic for research and policy, not just a marketing tactic. Future work should audit GEO in deployed systems and measure which concrete tactics actually change visibility and conversion. It should also develop defenses against optimization-driven bias in retrieval and ranking, and support clearer standards for provenance, disclosure, robustness, and fairness in LLM-mediated search and recommendation.

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